PyGame Tractor Pong Tutorial - Part 7

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Time required: 30 minutes

Collision time

1. Save **tractor_pong_6.py** as **tractor_pong_7.py**

```python
class TractorPong:
    # ------------------ INITIALIZE PYGAME ------------------------------#
    def __init__(self):
        # Pre initalize mixer with larger buffer size for better performance
        pygame.mixer.pre_init(
            44100,  # frequency (Hz)
            16,     # bit depth
            2,      # number of channels, 1 mono, 2 stereo
            4096    # buffer size, larger to optimize music playback.
        )

        # Initialize the pygame library
        pygame.init()

        # Create the game surface (window)
        self.surface = pygame.display.set_mode(
            (config.WIDTH, config.HEIGHT)
        )

        # Set window caption
        pygame.display.set_caption("Tractor Pong")

        # CLOCK object manages how fast the game runs
        self.clock = pygame.time.Clock()

        self.load_assets()
```

```python
# ---------------------- LOAD ASSETS -------------------------------------#
    def load_assets(self):
        # Load png image, use as program icon
        self.ball_ico = pygame.image.load(
            "./assets/blue_ball.png").convert_alpha()
        pygame.display.set_icon(self.ball_ico)

        # Load the images from the file system into a variable
        self.ball = pygame.image.load(
            "assets/soccer_ball.png").convert_alpha()
        self.tractor = pygame.image.load(
            "assets/green_tractor.png").convert_alpha()

        # Create a rectangle the same size as the image
        # rect is used to set the location of the image
        self.ball_rect = self.ball.get_rect()
        self.tractor_rect = self.tractor.get_rect()

        # Initial postion of the ball rectangle x random, y/top = 10
        self.set_ball_location()
        self.ball_rect.y = 10

        # Ball speed in pixels for x, y
        self.set_ball_direction()
        self.speed_y = 3
```

```python
        # Initial location of the tractor
        self.tractor_rect.left = config.WIDTH // 2
        self.tractor_rect.top = config.HEIGHT - 90

        # Speed in pixels for the tractor
        self.tractor_speed = 4

        # Keep track of score
        self.score = 0

        self.ball_hit = pygame.mixer.Sound("./assets/whip.wav")
        self.game_over_snd = pygame.mixer.Sound(
            "./assets/tractor_driving_game_over.wav"
        )
        # Set volume for sound effect in range 0.0 - 1.0
        pygame.mixer.Sound.set_volume(self.game_over_snd, .5)

        # Load and play background music
        pygame.mixer.music.load("./assets/tractor_driving.wav")

        # Set volume to 30%, range from 0.0 (mute) to 1.0 (full volume)
        pygame.mixer.music.set_volume(0.3)

        # Stop any other music from playing
        pygame.mixer.stop()

        # Play background game music in continious loop from the beginning
        pygame.mixer.music.play(-1)

        # Create font for scoring
        self.font_score = pygame.font.SysFont("Verdana", 20)

        # Only allow these events to be captured
        # This helps optimize the game for slower computers
        pygame.event.set_allowed(
            [
                pygame.QUIT,
                pygame.KEYDOWN,
                pygame.KEYUP,
            ]
        )
```

```python
# ------------------------- DISPLAY GAME OVER ----------------------------
    def game_over(self):
        """Display game over on top of the stopped game"""
        # Stop background sound
        pygame.mixer.music.stop()

        # Play game_over music until the user clicks a button
        pygame.mixer.Sound.play(self.game_over_snd, loops=-1)

        # Define a menu object for the game over screen
        game_over = pm.Menu(
            title="Game over",      # Set title menu to "Game over"
            width=config.WIDTH,     # Set to width of game surface
            height=config.HEIGHT,   # Set to height of game surface
            # Set the theme of the menu to an orange color scheme
            theme=pm.themes.THEME_ORANGE
        )

        # Display final score
        game_over.add.label(f"Score: {self.score}")

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Play Again?",    # Button text
            action=main             # Call main() to start over
        )

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Exit",           # Button text
            action=pm.events.EXIT   # Exit the game when clicked
        )

        # Run the main loop of the game over menu on the specified surface
        game_over.mainloop(self.surface)
```

Modify the update_ball method. If the ball hits the bottom, game over.

```python
# ------------------------- UPDATE BALL ----------------------------------#
    def update_ball(self):
        # Check for collision with left or right wall
        if self.ball_rect.left <= 0 or self.ball_rect.right >= config.WIDTH:
            # Reverse x direction multiply by -1
            self.speed_x = self.speed_x * -1

        # Check for collision with top or bottom wall
        if self.ball_rect.top <= 0 or self.ball_rect.bottom >= config.HEIGHT:
            # Reverse y direction multiply by -1
            self.speed_y = self.speed_y * -1

        # Move the ball position every frame
        self.ball_rect.x = self.ball_rect.x + self.speed_x
        self.ball_rect.y = self.ball_rect.y + self.speed_y

        # Ball hits bottom, player loses
        if self.ball_rect.bottom > config.HEIGHT:
            self.game_over()
```

The check collision method is changed.

```python
# ---------------------- CHECK COLLISION ----------------------------------#
    def check_collision(self):
        """Check for collision between two rects"""
        # The ball has to be above the tractor to collide
        # Does the ball collide with the tractor?
        # If so, reverse the ball y direction [1]
        if self.tractor_rect.colliderect(
            self.ball_rect
        ) and self.ball_rect.bottom < self.tractor_rect.top + 4:

            # Reverse y direction
            self.speed_y = self.speed_y * -1

            # Randomly change x direction
            direction = randint(0, 1)
            if direction == 0:
                self.speed_x = self.speed_x * -1

            # Increase speed by 10% each time the ball is hit
            self.speed_x = self.speed_x * 1.05
            self.speed_y = self.speed_y * 1.05

            # Increase score by 1
            self.score = self.score + 1
            pygame.mixer.Sound.play(self.ball_hit)
```

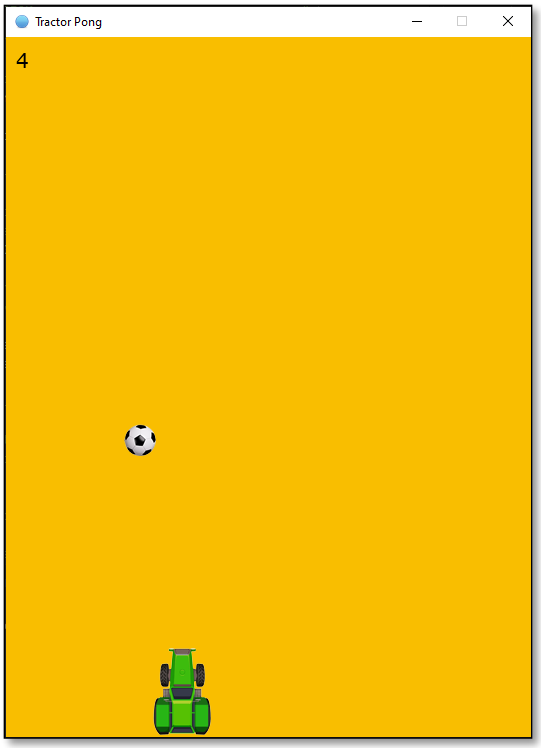
Draw the score on the screen.

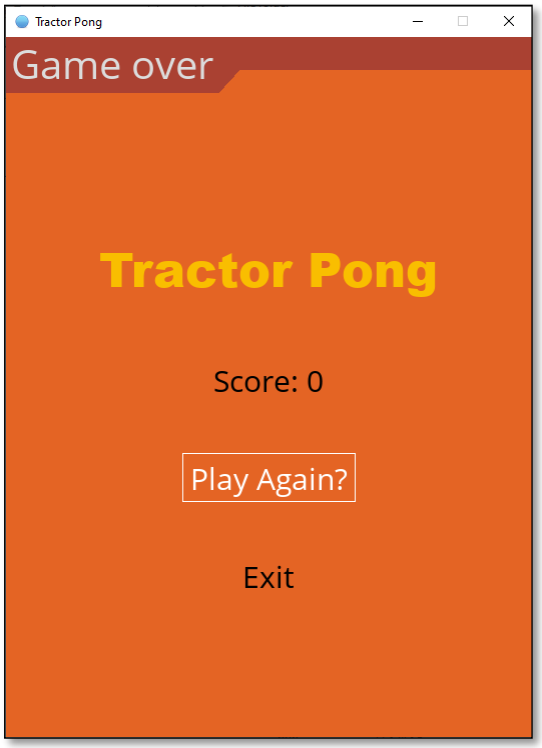
```python
# --------------------------- DRAW ----------------------------------------#
    def draw(self):
        # Fill the display surface to clear the previous screen
        # Comment out this line to see why is is necessary
        self.surface.fill(config.COUGAR_GOLD)

        # Draw the ball on the backbuffer
        self.surface.blit(
            self.ball,          # Image to draw
            self.ball_rect      # Location to draw the image
        )

        # Draw the tractor on the backbuffer
        self.surface.blit(
            self.tractor,       # Image to draw
            self.tractor_rect   # Location to draw the image
        )

        # Render score before drawing it on the surface
        score_display = self.font_score.render(
            f"{self.score}",    # Score
            True,               # Antialiasing true
            "black"             # Font color
        )

        # Draw score on the surface
        self.surface.blit(score_display, (10, 10))

        # ------- COPY BACKBUFFER INTO VIDEO MEMORY -----------------------#
        # Copy the backbuffer into video memory
        pygame.display.update()

```

Add a main method to allow the game to start over.

```python
def main():
    # Initialize program object and start game
    tractor_pong = TractorPong()
    tractor_pong.game_loop()


main()
```

Example run:





The tractor is King.

There is always room for improvement.

1. Add more difficulty levels.
2. Keep track of the highest score between games.
3. Add more music.
4. Change out the images.
5. Etc.

Assignment Submission

1. Attach all tutorials and assignments.
2. Attach screenshots showing the successful operation of each tutorial program.
3. Submit in Blackboard.